

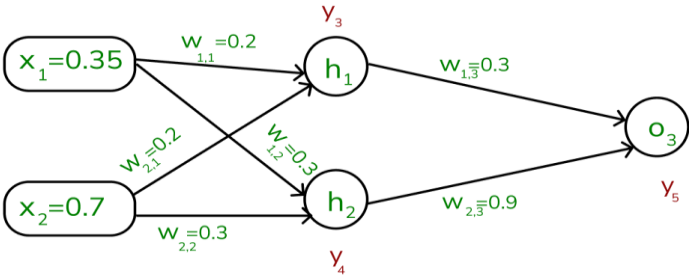


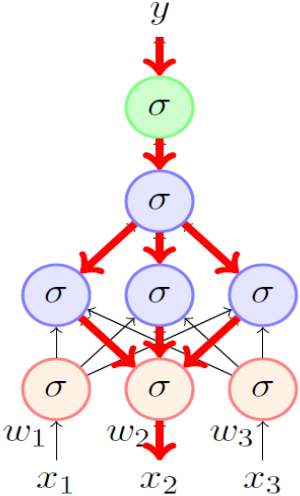
NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science & Engineering
Examination	End Semester
Program	B.Tech/PhD
Session	Spring
Course code & Course Name	CS6510 (Deep Learning)
Marks	50
Duration	3 hours

There are ... 5 ... questions printed on ...02... pages

Instructions: 1. In case of any doubt, write your assumptions, write it clearly and continue.
2. There are 05 questions printed on two pages of the paper. Attempt all the questions.

Q. No.	Questions	Marks	CO*	BL**
1. (A)	Given kernel of dimension $(k \times k)$ with width F , applied in an image with dimension $(W1, H1)$. Derive the equation for finding $(W2, H2)$ assuming that there are S strides and P padding during the first convolution operation. If there are K number of kernels in the ANN, what will be the number of parameters formed at the same time.	[5]	CO4 , CO2	Remembering (BL6) Applying (BL4)
(B)	Using back-propagation for neural networks, compute the output of nodes y_3, y_4, y_5 . Assume the activation function to be a sigmoid. Data: #Epochs = 2; Target output = 0.37; Learning rate = 0.01; Loss function = Absolute error; 	[5]		
2. (A)	Given a DNN with $ W $ weights, upon applying the dropout mechanism during training, the probability of deactivating weights is p . Assuming that there are m epochs, provide an algorithm/pseudo-code which decides an optimal set of W , so that the training of DNN has the fastest convergence time.	[5]	CO3, CO2	Understanding (BL5) Remembering (BL6)
(B)	Differentiate between the following variants of gradient descent (GD) methods i) Stochastic GD ii) Momentum GD iii) Nesterov Accelerated GD	[5]		

3. (A) (B)	Derive the generic term to calculate $grad(w_i)$ from the below given ANN. Subsequently find the expressions for $grad(w1)$, $grad(w2)$, $grad(w3)$.  Explain the <i>bias-variance</i> trade-off with appropriate diagram? Derive the equation for predicted error in terms of <i>bias</i> and <i>variance</i>.	[5] [5]	CO3, CO1	Understanding (BL5) Analyzing (BL3)
4. (A) (B)	Explain the complete architecture (with diagrams) of AlexNet model with appropriate demonstration of the convolution (CNV) and fully-connected (FC) layers. Compare and contrast with the LeNet-5 architecture. Given that the distribution of CNV and FC layers in the AlexNet Deep Neural Network (DNN) is $X : Y$. Suppose the altered distribution of these layers is changed to $(X-1):(Y+1)$. What impact will it have on the overall training process of the DNN.	[5] [5]	CO4, CO5	Understanding (BL5) Applying (BL4) Analyzing (BL3)
5. (A) (B)	Explain the complete architecture (with diagrams) of VGG16 model with appropriate demonstration of the CNV and FC layers. Compared to AlexNet and ResNet50, which of these above architectures would you prefer for image detection? Justify your answer with three concrete reasons. Given a DNN with number of layers $N = X + Y$, where $X = 0.4 CNV$ and $Y = FC$, out of X, Y which is more likely to suffer from the problem of vanishing gradient (VGD). Suggest a mechanism to tackle this problem of VGD with suitable mathematical equations.	[5] [5]	CO5, CO3	Understanding (BL5) Applying (BL4) Creativity (BL1)

*CO: Course Outcome; **BL: Bloom's Taxonomy Level.